



MOOGENE MEDI

Disclaimer

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2016	(주)무진메디 설립, 아주대학교 교원창업
2018	엔젤 / 시리즈 A 투자 유치 (50억원)
2020	브릿지 투자 유치 (30억원), 코넥스 상장
2022	탈모치료제 미국, 일본, 호주 등록 / 암 치료제 미국, 중국, 홍콩 특허 등록 아기유니콘 200 선정
2023	시리즈 B 투자 유치 (115억원)
2024	제약사와 mRNA 암백신 치료제 개발 공동연구
2025	탈모 치료제 IND 승인 (KFDA) 및 임상 펀딩 완료
2026	탈모치료제 임상1/2상, 코스메틱 사업 확장, 바이오원부자재 사업 선정

기업개요

일반현황

회 사 명	(주)무진메디
설 립 일	2016. 07. 06
자 본 금	63억원 (1,250만주, 액면가 500원)
임직원 수	23명 (2026년 9월)
주 소	하남 본사: 경기도 하남시 미사강변중앙로 7번안길 25, 유테크밸리 C동 10층 분당연구소: 경기도 성남시 분당구 정자일로 23 3층

대표이사



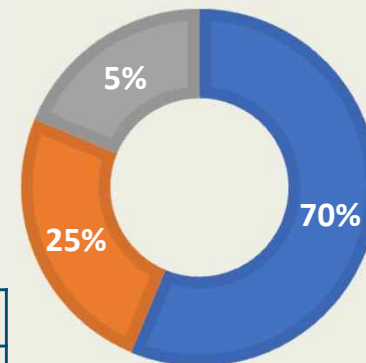
아주대 약대 교수
글로벌 제약 임상대학원장
주식회사 무진메디 대표

한국약제학회 임원
산자부 기획위원
농기평 기획위원

나노-바이오 연구 24년
SCI급 논문 84편
국가 연구 32건 수행

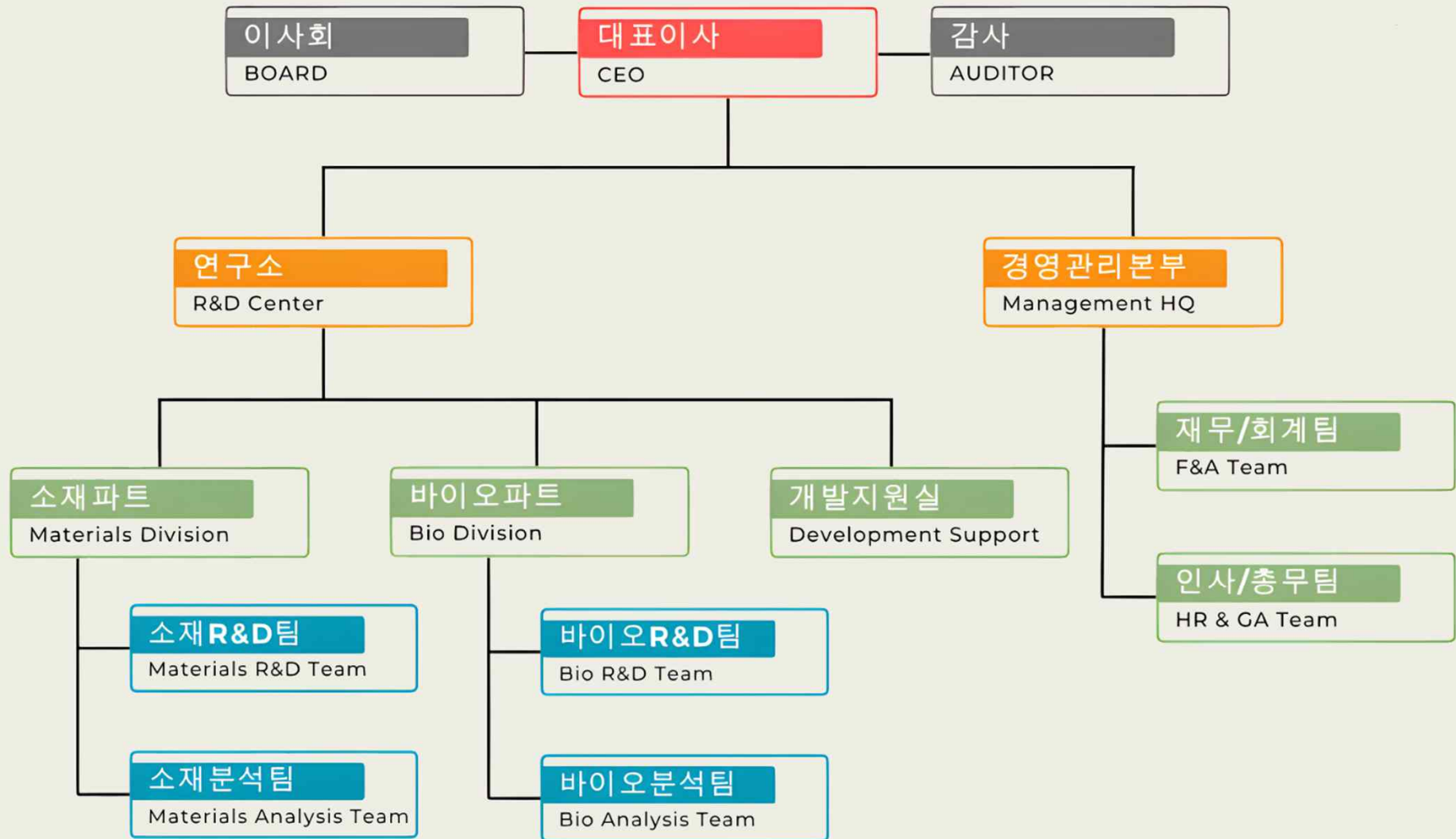
임직원 구성

- 연구개발
- 임원
- 관리



박사	5 (25 %)
석사	10 (45 %)
학사	8 (30 %)

기업개요

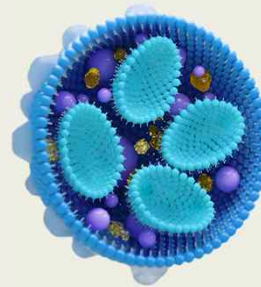




소수성
chemicals



mRNA



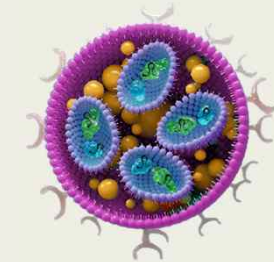
LNP

화장품 원료 개발
(기능성 화장품 /의료기기)

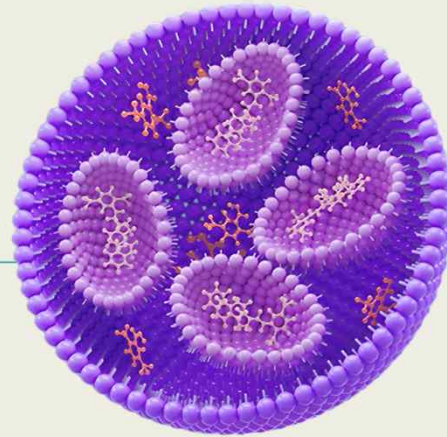
단백질



유전자 가위



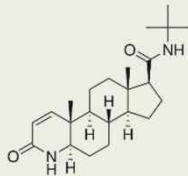
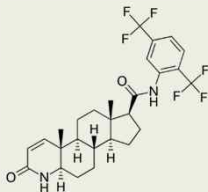
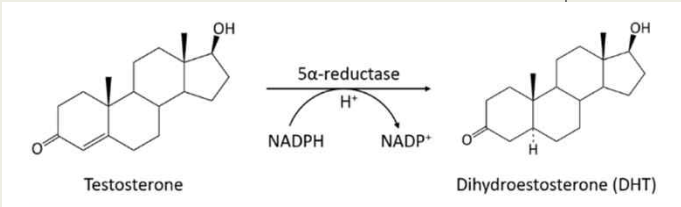
탈모 치료제 개발 현황



LNP-Dutasteride 제형
(약물의 부작용 극복)



피나스테리드 vs. 두타스테리드

제품명	Finasteride (프로페시아)	Dutasteride (아보다트)
용법/용량	 1mg / daily oral	 0.5mg / daily oral
MoA	 5-알파 환원효소 차단 → DHT 감소	
	1형 5-알파 환원효소 차단	1형, 2형 5-알파 환원효소 차단
반감기	8시간 이내	2주 이상 (~ 4주), 지속성에 의한 높은 효과
분포도 / 시장	1형 환원효소 분포: 피부, 간, 뇌, 땀샘, 전립선 글로벌 판매 중	1형 환원효소: 피부, 간, 뇌, 땀샘, 전립선 2형 환원효소: 피부, 전립선, 정낭, 고환, 간, 신장 탈모치료제: 한국, 일본, 대만 허가 글로벌: 전립선 비대증 치료제 허가



두타스테리드 외용제 개발 특징

장점

1. 국소 적용으로 부작용 최소화
2. 효과적인 DHT 차단: 긴 반감기를 통해 높은 농도의 환원효소 억제 효과
탈모 방지 지연에 큰 도움 될 것

도전 과제 (임상시험 필요)

1. 약물 흡수와 전달: 두피 흡수 효율 높이기 위한 제형 개발 필요
2. 국소 부작용: 두피 자극, 피부 건조, 발진 등의 국소 부작용 확인 필요
3. 장기 안전성: 장기적인 안전성을 확인을 위한 임상 시험 필요

세계 최초 두타스테리드 외용제 개발

두타스테리드 외용제 개발 (전문의약품)



✓ 주 성 분 : Dutasteride 0.25 mg/mL

✓ 제 형 : 국소 도포용 현탁액

✓ 효능 효과 : 남성형 탈모

✓ 용법 용량 : 탈모부위에 1일 1회 도포

(1~2번 펌프 하여 용기에 금속 볼로 문지름, 150~300 μ l)

1일 1회 펌핑시 최대 60일, 2회 펌핑시 최대 30일 사용 가능

2025년 8월 KFDA IND 승인

제약사 4개사 국내 공동개발 계약 완료

임상1/2 상 진행중 (서울대 병원)

12월 완료 예정

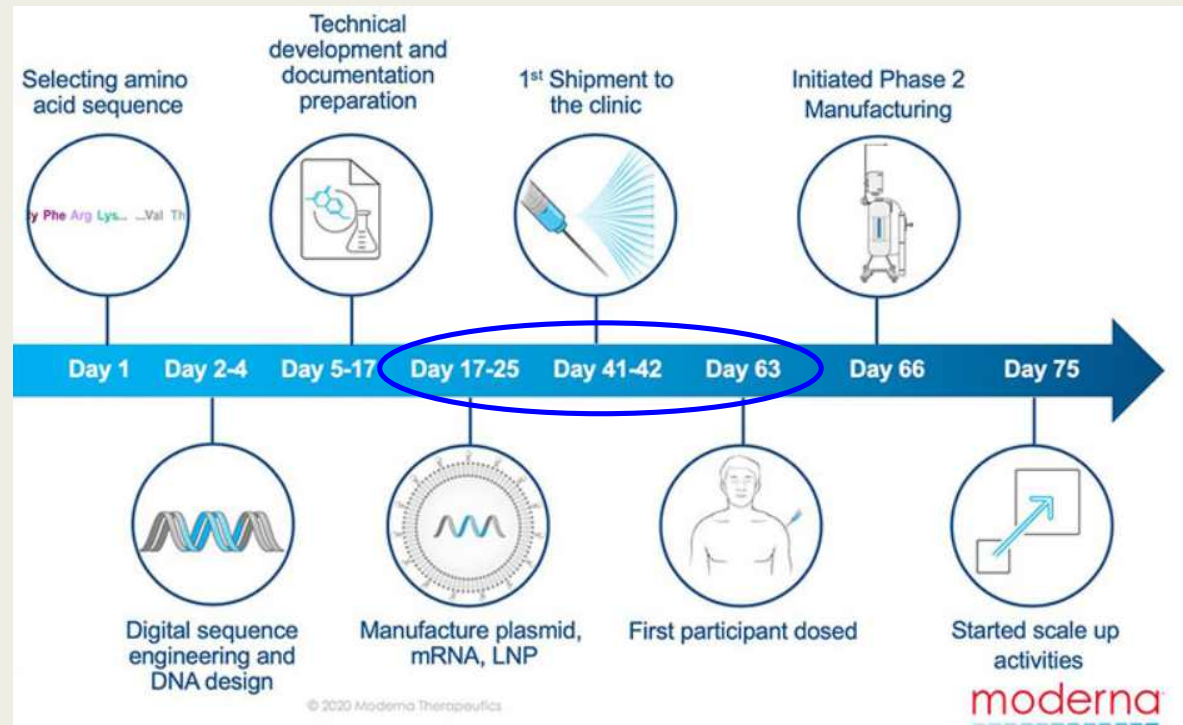


두타스테리드 외용제 개발 요약

항 목	내 용	비 고
개발품목	세계 최초 도포형 두타스테리드	전문 의약품
계약형태	공동개발계약	참 여 사 : 4개사 모집 완료 애드파마 : 임상 및 허가수행
계약기간	제품 발매 시점으로부터 10년간	
기술료	마일스톤 기반 수령	4개사 공동개발
예상보험약가	비급여	약국판매가 기준 약 4~5만원 예상 (최소 3개월 사용)
개발전략	임상1/2a, 임상3상 진행	서울대 병원 임상 시험 센터
임상1/2a	임상 대상자 투약 완료	결과 분석 진행중
출시예상	2028년 4분기	

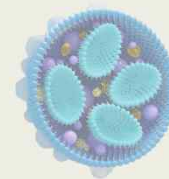
mRNA-LNP 개발 현황

환자 맞춤형 백신 (모더나 개발 중)



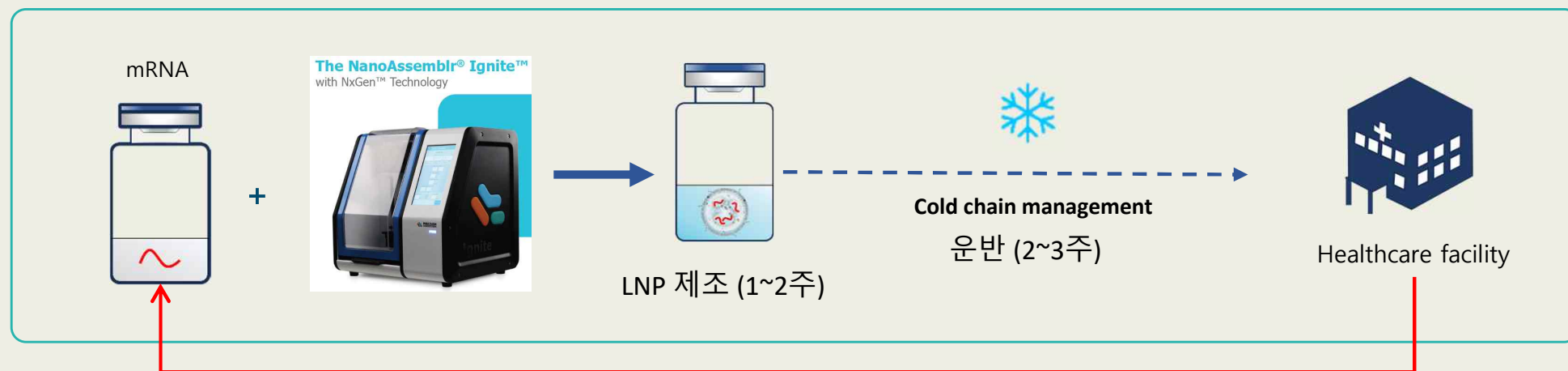
“needle to needle” time. 60일 이내로 줄이는 것이 최종 목표

환자 맞춤형 mRNA 확보 이후,
LNP 제형 제조 및 운반의 소요 기간이 문제임

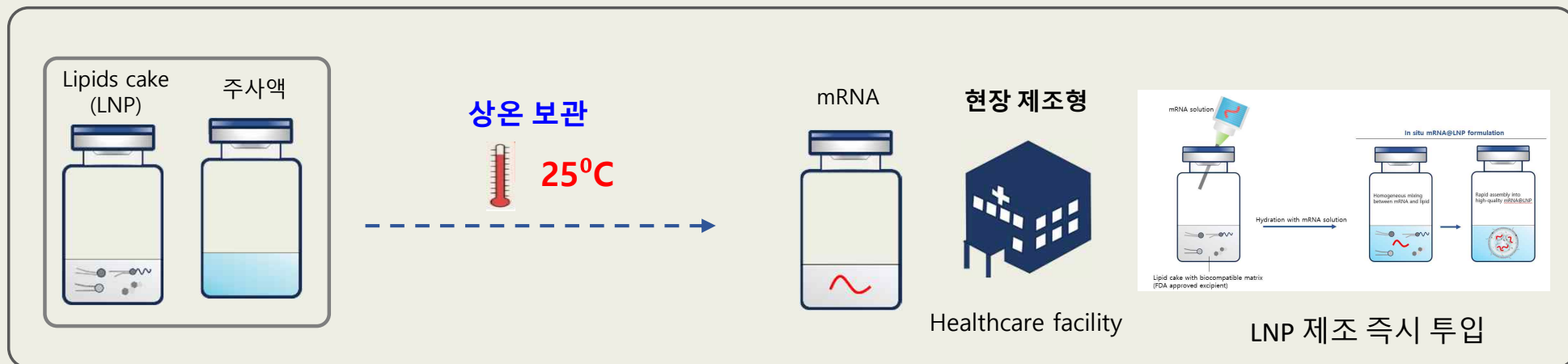


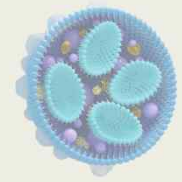
LNP – mRNA 치료제 플랫폼 개발

기존 기술 (mRNA 백신 LNP): 고가 장비 사용 및 보관/운반의 어려움



새로운 기술 (현장 제조형 mRNA 백신 LNP): 저가 생산비용 및 상온 보관/운반 가능





LNP – mRNA 치료제 플랫폼 개발

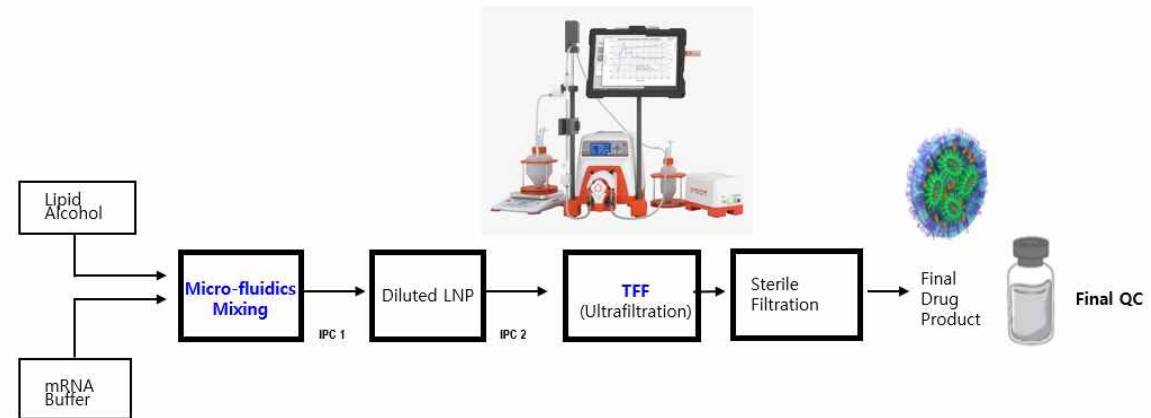
Pfizer Covid-19 Vaccine mRNA/LNP 제조

16개 Mixing 장비의 병렬조합을 통해 대량생산 (수억 dose)



The New York Times
"How Pfizer Makes Its Covid-19 Vaccine"

Process workflow for mRNA/LNP formulation

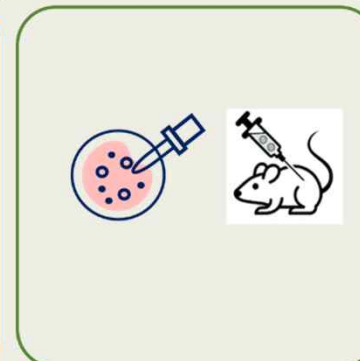
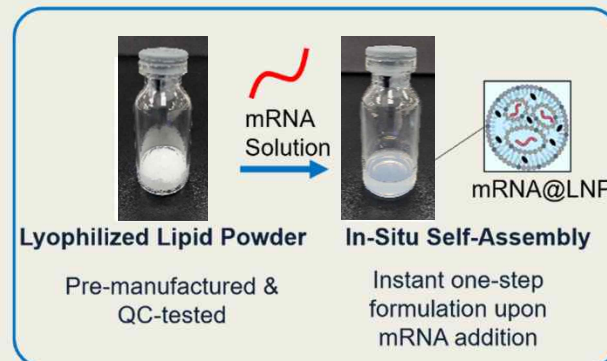


고가의 복잡한 제조 과정 필요

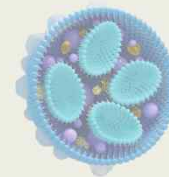


[1] Mix (Formulation)

[2] Go (Treatment)



동결건조기 제조 방법
(저가 생산 비용)

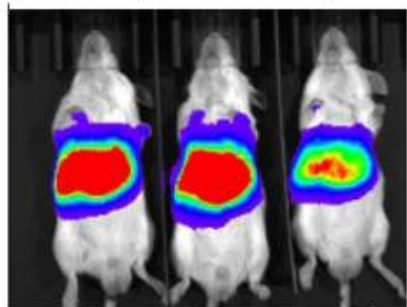


무진메디 mRNA-LNP 플랫폼

Target organ-specific delivery

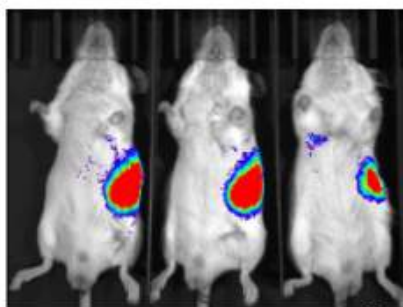
<Fluc mRNA@LNP 정맥 주사 후 발현 양상>

바이오엔텍 COVID-19 백신LNP

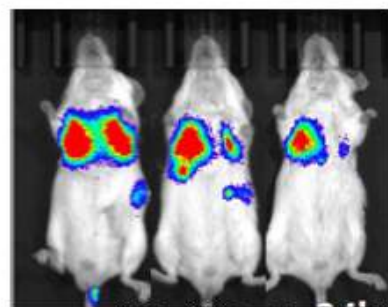


주요 장기: 간

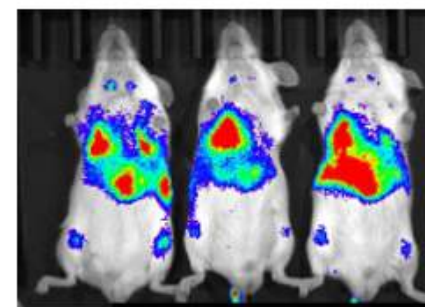
무진메디 LNP



주요 장기: 비장



주요 장기: 폐



주요 장기: 림프구, 폐, 간

mRNA를 전달하고자 하는 장기에 선택적인 LNP

질환에 따른 mRNA 전달용 LNP 개발 가능



Moogene's Cosmetics

코스메틱 분야의 현재 상황

1. 기능성 성분의 보편화 → 제품의 차별화 어려움
2. 낮은 침투도 → 제품 신뢰도 하락
3. 지용성 성분의 한계 → 알코올 사용, 제품의 다양화 어려움

무진 기술 요약

1. 기능성 성분의 LNP 제형 개발
2. 피부 침투도 **3배 증가** → 효능 증대
3. 지용성 성분의 **수용화** 가능 → 제품 다양화 (**무알코올**)

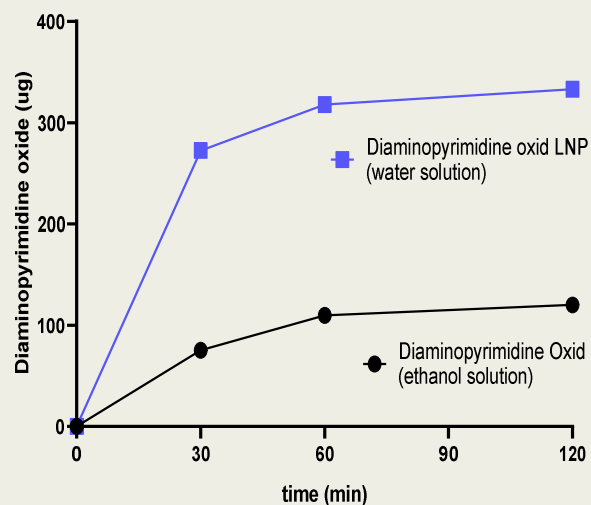
무진메디 코스메틱 원료 리스트

	Name	Active ingredients (INCI name)	Functions	Use
1	DAO LNP	Diaminopyrimidine Oxide	Diaminopyrimidine oxide, a minoxidil derivative, acts as a hair growth promoter to increase hair growth by acting on the hair follicles in the growth phase, and also expands peripheral capillaries to promote the nutrients necessary for hair growth and proliferation of hair papilla cells.	Hair tonic/essence
2	AHP8 LNP	Acetyl hexapeptide-8	It is known as a botox-like peptide and is primarily used in cosmetics for wrinkle reduction and skin firming benefits	Wrinkle Cream/serum /mask pack
3	Cera LNP	Ceramide-NP	It is a ceramide-based lipid ingredient with excellent functions in strengthening the skin barrier and retaining moisture. In particular, it is a long-chain ceramide known for its high stability and long-lasting moisturizing effect.	Regeneration Cream/serum /essence
4	NA LNP	Niacinamide	It is an amide form of vitamin B3 and one of the most extensively studied skincare ingredients, recognized for its wide range of benefits, including brightening, wrinkle improvement , anti-inflammatory effects, and skin barrier enhancement.	Brightening cream/serum /essence
5	DCA LNP	Deoxycholic acid	Harnessing FDA-approved deoxycholic acid in a next-generation LNP delivery system, this formulation allows for effective fat-dissolving action-without the need for injections, Experience non-invasive contouring with clinical trusted results.	Slimming Cream/patch
6	Phyto LNP	Phytosphingosine	It is a natural sphingoid base known for its diverse skin benefits, including skin barrier protection, anti-inflammatory and antibacterial effects, and acne improvement .	Anti-inflammation Acne Cream/serum /essence
7	Tranexamic acid LNP	Tranexamic acid	With proven melanin-inhibiting properties, this ingredient is effective in reducing melasma , evening out skin tone, and soothing redness for a clearer, more radiant complexion.	Anti-Melasma Cream/serum
8	4-Butyl resorcinol LNP	4-Butyl resorcinol	Helps reduce the appearance of melasma and age spots by blocking melanin production, the key pigment responsible for uneven skin tone.	Anti-Melasma Cream/serum
9	α -tocopherol LNP	α -tocopherol	A powerful antioxidant form of vitamin E that defends against free radicals, refines skin texture, and enhances hydration for a healthier-looking complexion.	Anti-aging Sun block/balm Cream
10	β -sitosterol LNP	β -sitosterol	This plant-based cholesterol (Phytosterol) helps brighten the skin, calm irritation, and even promote healthier hair growth .	Hair tonic/essence
11	Bakuchiol LNP	Bakuchiol	A gentle, plant-based alternative to vitamin A retinol , offering powerful anti-aging and skin-repair benefits. It visibly smooths wrinkles, enhances elasticity, and evens out skin tone – all without the irritation commonly associated with retinoids.	Anti-aging /brightening Cream/serum /essence



탈모 헤어 토닉: 피부 침투도 및 효능

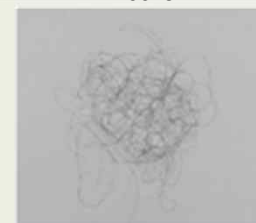
Skin penetration test (Franz cell test)



Baseline



4 weeks



8 weeks



Table 3. Statistical analysis of shedding hair count

Time point	N	Mean ¹	SD	p-value ²	Change from baseline (%)
Baseline	11	82.73	56.98	-	-
4 weeks	11	49.00	30.25	0.067	40.77▼
8 weeks	11	39.45	23.43	0.034*	52.31▼

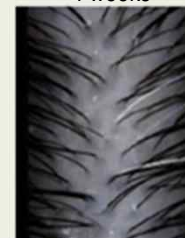
¹Decrement of the mean value represents decrease of shedding hair count (▼).

²Significantly different at * $p < 0.05$ compared with baseline

Baseline



4 weeks



8 weeks

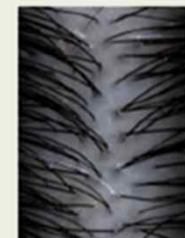


Table 5. Statistical analysis of hair density grading

Time point	N	Mean ¹	SD	p-value ²
Baseline	11	0.00	0.00	-
4 weeks	11	0.64	0.39	0.006* ^W
8 weeks	11	0.86	0.32	0.003* ^W

¹Increment of the mean value represents improvement of hair density.

²Significantly different at * $p < 0.025$ compared with baseline

^WWilcoxon Signed-Ranks Test (Bonferroni correction)



탈모 헤어 토닉: 요약 설명



Powerful LNP technology

When LNPs are used as carriers for the delivery of active ingredients, they provide multiple functions: in addition to the intrinsic effects of the active ingredients, LNPs can enhance penetration, improve solubility and increase stability of active molecules.

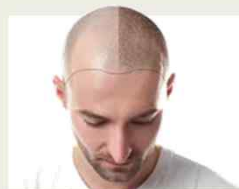


Natural scalp strengthening ingredients

Our clinically proven formula is crafted with ingredients sourced safely and ethically. The intense hair growth tonic is vegan-friendly and formulated without parabens, sulfates, silicones or phthalates.



Real results (only 8 weeks)



Reduction in the number of
hair loss

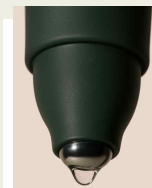
52.3%

Increased hair density

0.86 UP

Increased skin blood flow

6%

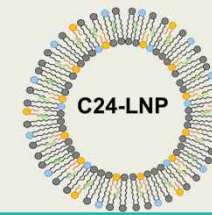


hand-free applicator

Roll on airless pump bottle dispenses a controlled amount of product without getting it on your head.

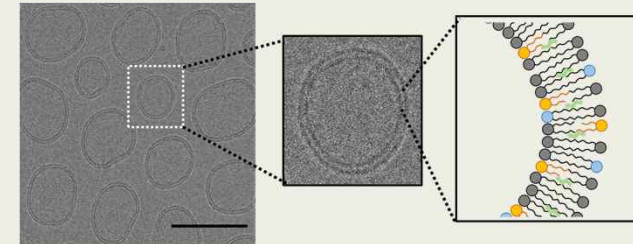
- "Airless technology" prevents air exposure and contamination, preserving the product's efficacy.
- "Roll on applicator" provides a gentle, cooling massage during application, promoting circulation and reducing puffiness

세라마이드 LNP (재생 효능)



Ceramide-24 LNP: Nano carrier system

무진메디 고유 기술을 통한 지용성 화합물의 수용화



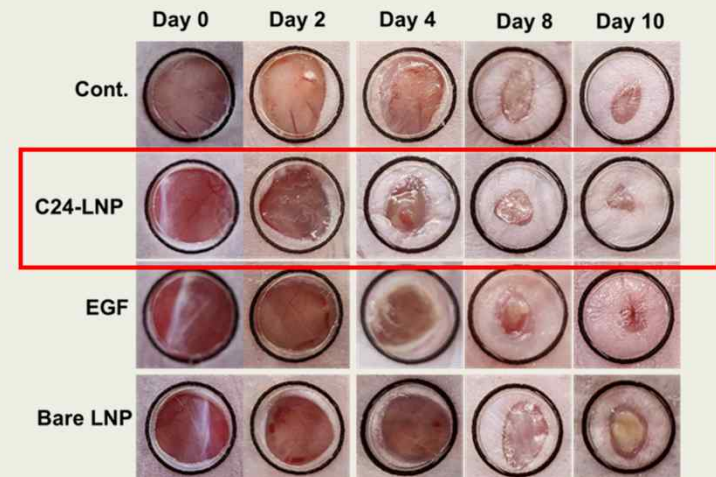
세포 성장 촉진

세포 성장을 140 % 촉진, 약물 없이도 피부 세포 성장을 급속히 성장

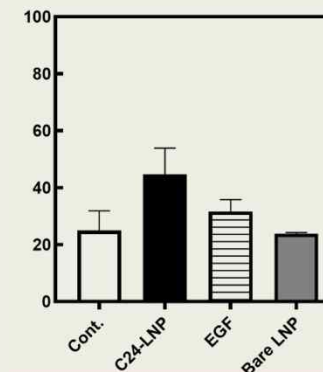
특허와 논문으로 검증된 기술

세포 성장 기능 관련 특허 등록 완료

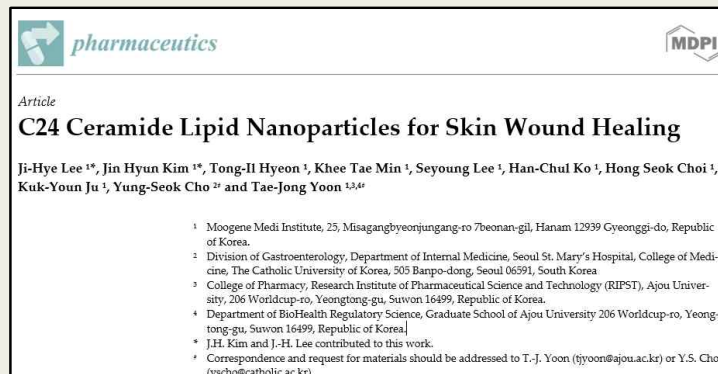
약물 관련 전문 논문에 게재 하여 검증 완료



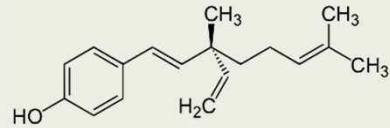
Day 4



단백질 바이오의약품인
EGF보다 높은 재생 효과를 보임



Bakuchiol LNP



- Efficacy in vivo

Atopic dermatitis model

G1 (염증유발x)



G2 (염증유발o)



G3 (DMT)



G4 (BKC1.5)



G5 (BKC0.5)



G6 (BKC0.1)



G7 (BKC-LNP1.5)



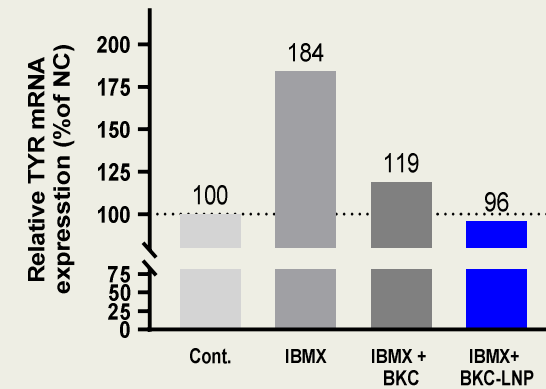
G8 (BKC-LNP0.5)



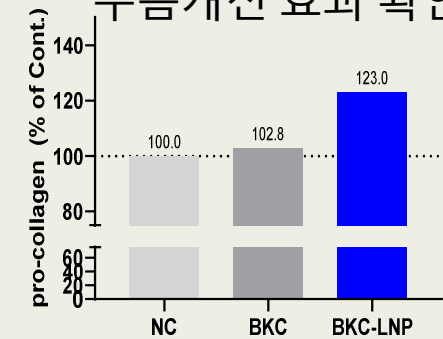
G9 (BKC-LNP0.1)



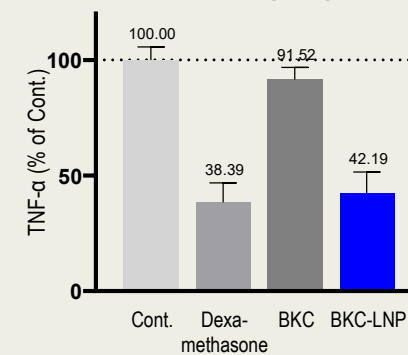
미백 효과 확인



주름개선 효과 확인



항염 효과 확인





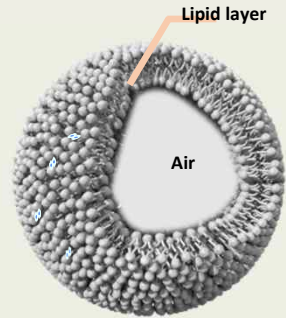
마이크로 버블 코스메틱 제품 개발



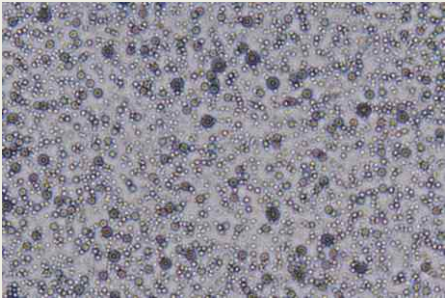
초음파 감응형 전달 기술

마이크로 버블 - 코스메틱 제품 개발

Microbubble

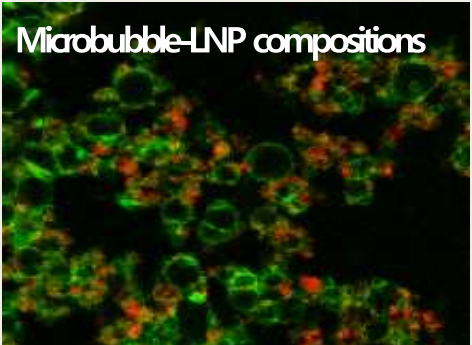


Microscopic image (Microbubble)



Ultrasound (1MHz, 0.3W 10s)

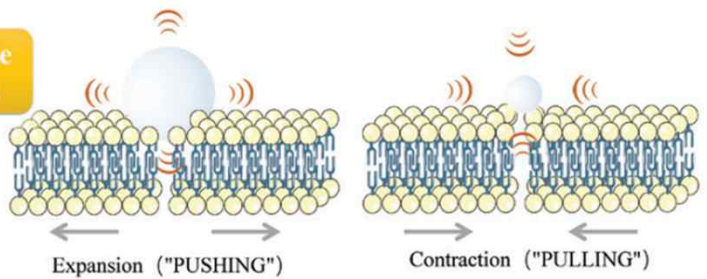
Microscopic image (After cavitation)



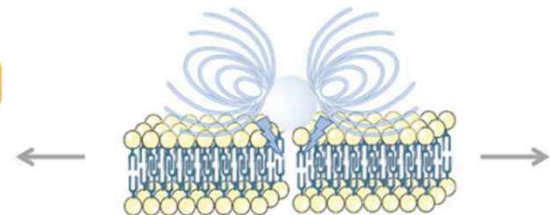
Theranostics 2014, Vol. 4, Issue 11
1133
IVYSPRING
INTERNATIONAL PUBLISHER
Theranostics
2014; 4(11): 1133-1144. doi: 10.7150/tno.9945
Short Research Communication
Ultrasound-Mediated Gene and Drug Delivery Using a Microbubble-Liposome Particle System
Young Il Yoon^{1,2*}, Yong-Su Kwon^{3*}, Hee-Sang Cho⁴, Sun-Hee Heo⁵, Kyeong Seom Park⁶, Sang Gyu Park⁶, Soo-Hong Lee⁶, Seung Il Hwang^{1,2}, Young Il Kim¹, Hwan Jun Jae¹, Gook-Jun Ahn⁷, Young-Seok Cho⁸, Hakho Lee^{6,9}, Hak Jong Lee^{1,2,3,10}, Tae-Jong Yoon^{1,10}

Biomaterials
Volume 232, February 2020, 119736
ELSEVIER
Ultrasound-activated particles as CRISPR/Cas9 delivery system for androgenic alopecia therapy
Jee-Yeon Ryu, Eun-Jeong Won, Han A Reum Lee, Jin Hyun Kim, Emmanuel Hui, Hong Pyo Kim, Tae-Jong Yoon & 8

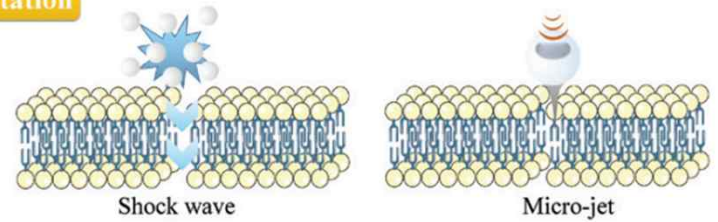
Microbubble oscillations



Microstreaming



Inertial cavitation



마이크로 버블 – 코스메틱 제품 개발

How to Use:

Microbubble Ampoule/Serum

Composition

Freeze dry cake (Mold)

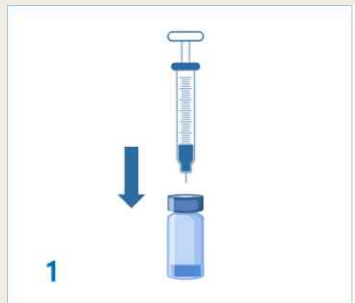
Hyaluronic acid mold

microbubble cake

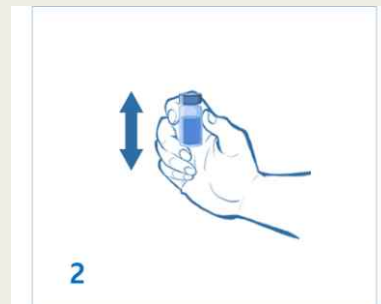


Aqueous solution for hydration

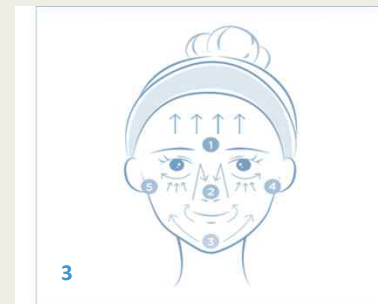
Active ingredients
(PDRN, Niacinamide, Peptide etc)



1. Open the freeze dry vial and add 2 ml of solution



2. Shake vigorously for 20 to 30 seconds.



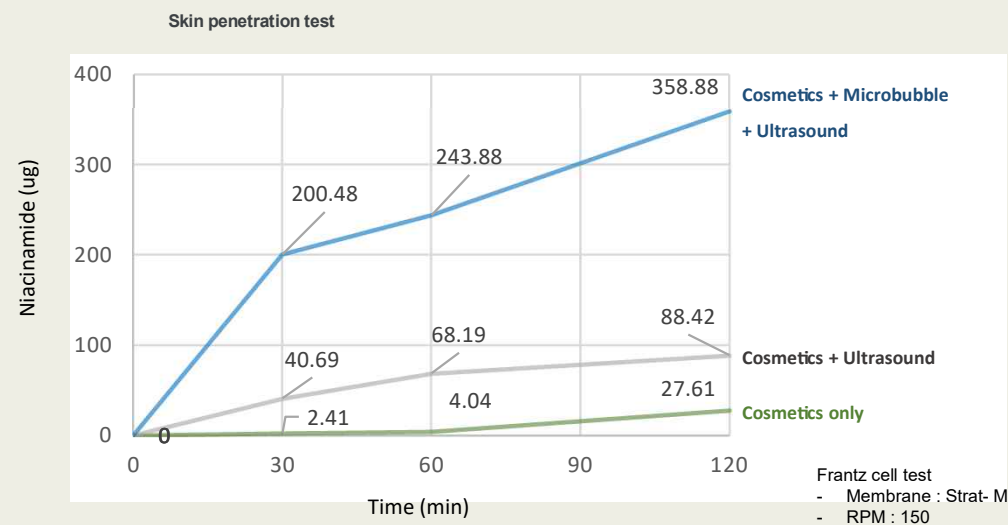
3. Apply to face



4. Massage with ultrasound immediately after application

마이크로 버블 - 코스메틱 제품 개발

- Efficacy test (in vitro)



“무궁무진한 나노-바이오 의약 기술의 선두주자 무진메디 ”



주식회사 무진메디

경기도 하남시 미사강변중앙로 7번안길 25, 유테크밸리 10층
T 031-8027-6150 F 031-8027-6159